

WEEK 4 – ASSIGNMENT WITH RESULTS

Title:

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

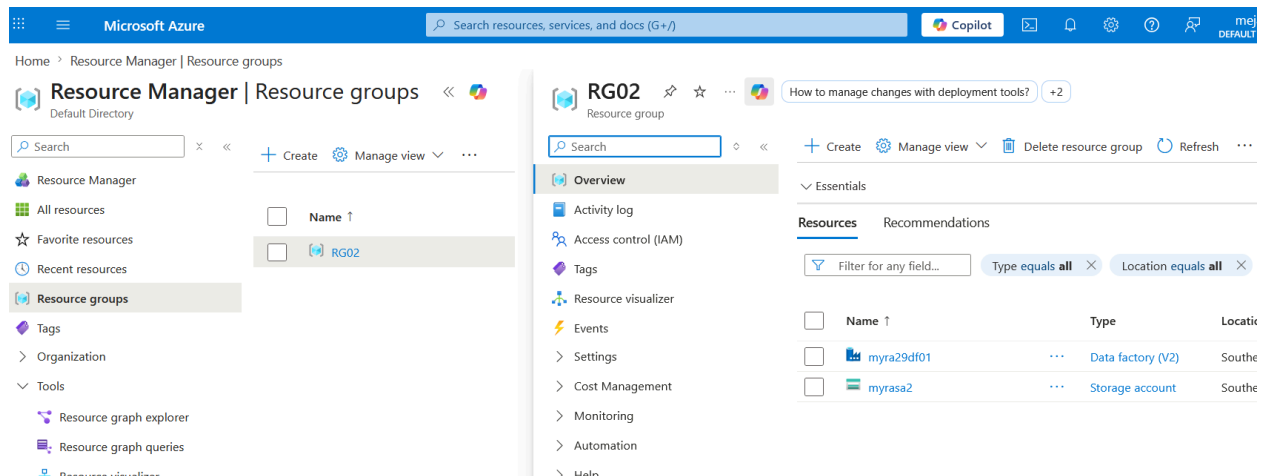
Assignment Tasks

Task 1:

- Explore Azure Portal
- Create a Resource Group

Deliverable:

- Screenshot of Resource Group



I made a resource group named RG02 (region : central india)

Task 2: Storage Setup

Do:

- Create a Storage Account

WEEK 4 - ASSIGNMENT

- Create a Blob Container
- Upload a CSV file

Deliverable:

- Screenshot of container with uploaded file

The top screenshot shows the Azure Storage portal interface for the 'myrasa2' storage account. The 'Containers' view is active, showing a list of containers. The table below represents the data shown in the screenshot:

Name	Last modified	Anonymous access level	Lease state	Storage connector
\$logs	27/6/2026, 12:44:37 pm	Private	Available	-
blobc01	27/6/2026, 12:47:19 pm	Private	Available	-

The bottom screenshot shows the 'blobc01' container view. The 'Overview' tab is selected, displaying the authentication method as 'Access key'. Below, a table lists the blobs uploaded to the container:

Name	Last modified	Access tier	Blob type	Size	Lease state
Sample - Superstore.csv	27/6/2026, 12:49:58 pm	Hot (Inferred)	Block blob	2.19 MiB	Available
destination.csv	27/6/2026, 2:06:15 pm	Hot (Inferred)	Block blob	2.19 MiB	Available

Made storage account 'myrasa2' (region which was able to verify was southeast asia)
Inside i made a blob container 'blobc01' and then uploaded 2 csv files for further adf pipeline tasks.

Task 3: ADF Basics

Do:

- Create Azure Data Factory
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)
- Create datasets (source + destination)
- Use Get Metadata activity

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Deliverable:

- Screenshot of Linked Service
- Screenshot of dataset
- Screenshot of Get Metadata activity

Data factory 'myra29df01' was created

Home

myra29df01 Data factory (V2)

Search Delete

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Settings
 - Networking
 - Managed identities
 - Properties
 - Locks
- Getting started
- Monitoring

Essentials

Resource group (move) : [RG02](#)

Status : Succeeded

Location : Southeast Asia

Subscription (move) : [Azure for Students](#)

Subscription ID : 6bb275d4-0369-4cc0-9da7-a512fbc31cdc

Type : Data factory (V2)

Getting started : [Quick start](#)



Azure Data Factory Studio

[Launch studio](#)

Blob container 'blobc01' was linked

Microsoft Azure | Data Factory | myra29df01

Search

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Migrate to Fabric (Preview)

Linked services

Linked service defines the connection information to a data store or compute. [Learn more](#)

+ New

Filter by name Annotations : Any

Showing 1 - 1 of 1 items

Name ↑↓	Type ↑↓	Related ↑↓	Annotations ↑↓
AzureBlobStorage1	Azure Blob Storage	2	

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Source file-superstore.csv

The screenshot shows the Microsoft Azure Data Factory portal. On the left, the 'Factory Resources' pane is expanded to 'Datasets', showing 'DelimitedText1' and 'DelimitedText2'. The main pane displays the configuration for 'DelimitedText1'. The 'Connection' tab is active, showing the following settings:

- Linked service: AzureBlobStorage1 (Test connection: Connection successful)
- File path: blobc01 / Directory / Sample - Superstore.csv (Browse button available)
- Compression type: No compression
- Column delimiter: Comma (,)
- Row delimiter: Default (\r\n, or \n)
- Encoding: Default(UTF-8)
- Quote character: Double quote (")
- Escape character: Backslash (\)

Destination file-destination.csv

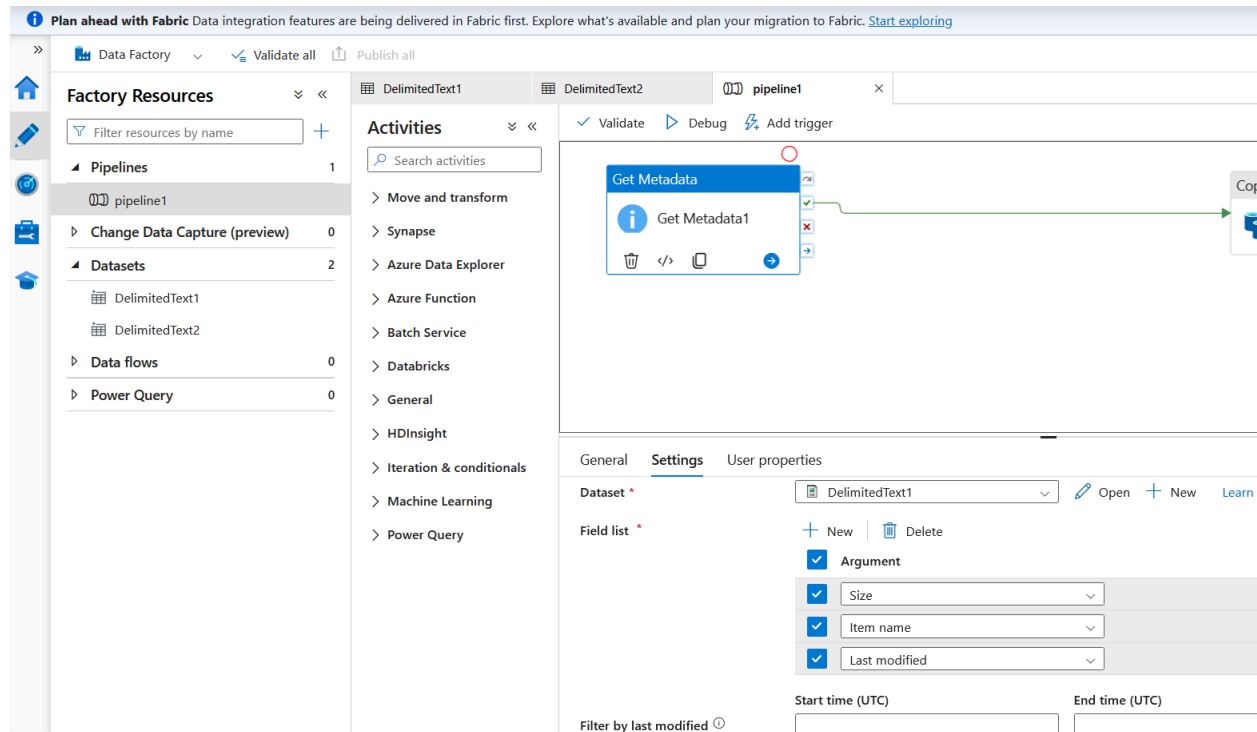
The screenshot shows the Microsoft Azure Data Factory portal. On the left, the 'Factory Resources' pane is expanded to 'Datasets', showing 'DelimitedText1' and 'DelimitedText2'. The main pane displays the configuration for 'DelimitedText2'. The 'Connection' tab is active, showing the following settings:

- Linked service: AzureBlobStorage1 (Test connection: Connection successful)
- File path: blobc01 / Directory / destination.csv (Browse button available)
- Compression type: No compression
- Column delimiter: Comma (,)
- Row delimiter: Default (\r\n, or \n)
- Encoding: Default(UTF-8)
- Quote character: Double quote (")
- Escape character: Backslash (\)
- First row as header: ☒

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Within the ADF framework, the **Get Metadata** activity allows for the extraction of property details regarding various storage, objects such as directories or individual files, enabling the retrieval of structural information without the necessity of processing or transferring the internal data payloads.

In our case we are getting details such as **size, item name , and last modified**



Task 4: Pipeline Development

Do:

- Create pipeline using Copy Data activity
- Configure source and destination
- Add ForEach activity (optional if multiple files)

Deliverable:

- Screenshot of pipeline design

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Within the ADF framework, the **Get Metadata** activity allows for the extraction of property details regarding various storage objects such as directories or individual files, enabling the retrieval of structural information without the necessity of processing or transferring the internal data payloads; conversely, the **Copy Data** activity functions to reliably ingest, convert, and relocate these datasets across diverse cloud environments.

Acting as a looping mechanism within Azure Data Factory (ADF), the **ForEach** activity iterates through an item collection—such as database tables or file names—to run a designated sequence of nested activities for every element in the array.

Note: we have not used foreach in our pipeline since we are not using multiple datasets .

The screenshot displays the Azure Data Factory (ADF) web interface. At the top, there are tabs for 'DelimitedText1', 'DelimitedText2', and 'pipeline1'. Below the tabs, there's a navigation bar with 'Activities' and a search bar. The 'Activities' panel on the left lists various activities under 'Databricks' and 'General'. The main canvas shows a pipeline named 'pipeline1' with two activities: 'Get Metadata' (labeled 'Get Metadata1') and 'Copy data' (labeled 'Copy data1'). A green arrow indicates the flow from 'Get Metadata' to 'Copy data'. The 'Copy data' activity is highlighted, and its configuration panel is visible at the bottom. The configuration panel has tabs for 'General', 'Source', 'Sink', 'Mapping', 'Settings', and 'User properties'. The 'Source' tab is active, showing 'Source dataset' as 'DelimitedText1' and 'File path type' as 'File path in dataset'. There are also options for 'Start time (UTC)' and 'End time (UTC)'.

Task 5: Pipeline Execution

Do:

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- Run pipeline using Debug/Trigger

Deliverable:

- Screenshot showing pipeline execution (Succeeded)

The screenshot displays the Azure Data Factory interface for a pipeline named 'pipeline1'. The pipeline consists of two activities: 'Get Metadata2' and 'Copy data1', both of which have completed successfully, as indicated by green checkmarks. The 'Output' tab is selected, showing the pipeline run ID 'c553d5e7-e5ec-4a3e-a90b-523457190965' and the status 'Succeeded'. A table below provides details for the activities:

Activity name	Activity status	Activity name	Run start	Duration	Integration runtime
Copy data1	Succeeded	Copy data	6/28/2026, 3:10:21 PM	16s	AutoResolveIntegrationRuntime (Southeast)
Get Metadata2	Succeeded	Get Metadata	6/28/2026, 3:10:08 PM	12s	AutoResolveIntegrationRuntime (Southeast)

Task 6: IAM Roles

Do:

- Assign roles:
 - Reader
 - Contributor
- Provide access to ADF for Storage

Deliverable:

- Screenshot of role assignment

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Microsoft Azure

mejam892@gmail.com
DEFAULT DIRECTORY (MEJAM89...

Home > myrasa2 | Access Control (IAM)

Add role assignment

Role

Members

Conditions

Review

Selected role

Reader

Assign access to

☐ User, group, or service principal

☒ Managed identity

Members

+ Select members

Name	Object ID
No members selected	

Description

Optional

Select managed identities

Some results might be hidden due to your ABAC condition.

Subscription *

Azure for Students

Managed identity

Data factory (V2) (1)

Search by name

Selected members:

 myra29df01
/subscriptions/6bb275d4-0369-4cc0-9da7-a512fbc31cdc/resourceGroups/...

Remove

Review + assign

Previous

Select

Close

Feedback

WEEK 4 - ASSIGNMENT

myrasa2 | Access Control (IAM)

Storage account

Search

◊ <<

+ Add ▾

↓ Download role assignments

≡ Edit columns

🔄 Refresh

⋮

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Containers

Classic file shares

Queues

Tables

Security + networking

Networking

Check access

Role assignments

Roles

Deny assignments

Number of role assignments for this subscription ⓘ

44000

Search by name or email

Type : All

Role : All

Scope : All scopes

Group by : Role

All (4)

Job function roles (2)

Privileged administrator roles (2)

☐

Name ↑↓

Type ↑↓

Role ↑↓

Scope ↑↓

Owner (2)

☐

Myra Jain

b095c7a2-5b0...

User

Owner

Subscription (

☐

Myra Jain

b095c7a2-5b0...

User

Owner

Subscription (

Reader (1)

☐

myra29df01

8072858e-5cd...

Managed identity

Reader

This resource

Storage File Data Privileged Contributor (1)

☐

myra29df01

8072858e-5cd...

Managed identity

Storage File Data Privil...

This resource

Mini Project (Final Task)

Problem Statement:

Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

Requirements:

- Source: CSV file in Blob
- Use: Linked Service + Dataset + Pipeline
- Process: Copy Data + Metadata check
- Destination: New file/location

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Expected Output:

- Pipeline executed successfully
- Data copied to destination
- Metadata validated

Home

myra29df01 Data factory (V2) [Summarize the properties of this Data factory \(V2\).](#) [List triggers associated with this Data factory \(V2\).](#) [Show me integration runtime metrics for this Data factory \(V2\).](#)

Delete

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Settings
 - Networking
 - Managed identities
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 - Locks
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- Monitoring

Essentials

Resource group [\(move\)](#) : RG02 Type : Data factory (V2)

Status : Succeeded Getting started : [Quick start](#)

Location : Southeast Asia

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Subscription ID : 6bb275d4-0369-4cc0-9da7-a512fbc31cdc



Azure Data Factory Studio

[Launch studio](#)

Microsoft Azure | Data Factory | myra29df01 mejam892@gmail.com

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Data Factory Validate all Publish all [Migrate to Fabric \(Preview\)](#)

General

- Connector upgrade advisor
- Factory settings

Connections

- Linked services**
- Integration runtimes
- Microsoft Purview
- ADF in Microsoft Fabric

Linked services

Linked service defines the connection information to a data store or compute. [Learn more](#)

[+ New](#)

Annotations : **Any**

Showing 1 - 1 of 1 items

Name ↑↓	Type ↑↓	Related ↑↓	Annotations ↑↓
AzureBlobStorage1	Azure Blob Storage	2	

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Factory Resources

- Pipelines: 1
- Change Data Capture (preview): 0
- Datasets: 2
 - DelimitedText1
 - DelimitedText2
- Data flows: 0
- Power Query: 0

DelimitedText1

DelimitedText1

Connection Schema Parameters

Linked service: AzureBlobStorage1 [Test connection](#) [Edit](#) [New](#) [Learn more](#)

File path: blobc01 / Directory / Sample - Superstore.csv [Browse](#) [Preview data](#) [Detect format](#)

Compression type: No compression

Column delimiter: Comma (,)

Row delimiter: Default (\r,\n, or \r\n)

Encoding: Default(UTF-8)

Quote character: Double quote (")

Escape character: Backslash (\)

Microsoft Azure | Data Factory | myra29df01 | Search | mejam892@gmail.com

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Factory Resources

- Pipelines: 1
- Change Data Capture (preview): 0
- Datasets: 2
 - DelimitedText1
 - DelimitedText2
- Data flows: 0
- Power Query: 0

DelimitedText1 DelimitedText2

DelimitedText2

Connection Schema Parameters

Linked service: AzureBlobStorage1 [Test connection](#) [Edit](#) [New](#) [Learn more](#)

File path: blobc01 / Directory / destination.csv [Browse](#) [Preview data](#)

Compression type: No compression

Column delimiter: Comma (,)

Row delimiter: Default (\r,\n, or \r\n)

Encoding: Default(UTF-8)

Quote character: Double quote (")

Escape character: Backslash (\)

First row as header: ☒

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Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Data Factory | Validate all | Publish all

Factory Resources

Filter resources by name

- Pipelines 1
 - pipeline1
- Change Data Capture (preview) 0
- Datasets 2
 - DelimitedText1
 - DelimitedText2
- Data flows 0
- Power Query 0

Activities

Search activities

- Move and transform
- Synapse
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

pipeline1

Validate | Debug | Add trigger

Get Metadata

Get Metadata1

General | **Settings** | User properties

Dataset * DelimitedText1 [Open](#) [New](#) [Learn](#)

Field list * [New](#) [Delete](#)

- ☒ Argument
- ☒ Size
- ☒ Item name
- ☒ Last modified

Start time (UTC) End time (UTC)

Filter by last modified ⓘ

DelimitedText1 | DelimitedText2 | **pipeline1**

Validate | Validate copy runtime | Debug | Add trigger

Activities

Search activities

- Databricks
- General
 - Append variable
 - Delete
 - Execute Pipeline
 - Execute SSIS package
 - Fail
 - Get Metadata
 - Lookup
 - Stored procedure

Get Metadata

Get Metadata1

Copy data

Copy data1

General | **Source** | Sink | Mapping | Settings | User properties

Source dataset * DelimitedText1 [Open](#) [New](#) [Preview data](#) [Learn more](#)

File path type ☒ File path in dataset ☐ Prefix ☐ Wildcard file path ☐ List of files ⓘ

Start time (UTC) End time (UTC)

WEEK 4 - ASSIGNMENT

DelimitedText1

DelimitedText2

pipeline1

Validate

Debug

Add trigger

Activities

Search activities

Databricks

General

Append variable

Delete

Execute Pipeline

Execute SSIS package

Fail

Get Metadata

Lookup

Stored procedure

Script

Set variable

Validation

Web

WebHook

Get Metadata

Copy data

Get Metadata2

Copy data1

Parameters

Variables

Settings

Output

Pipeline run ID

c553d5e7-e5ec-4a3e-a90b-523457190965

Pipeline status

Succeeded

View debug run consumption

All status

Monitor in Azure Metrics

Export to CSV

Showing 1 - 2 of 2 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Copy data1	Succeeded	Copy data	6/28/2026, 3:10:21 PM	16s	AutoResolveIntegrationRuntime (Southeast
Get Metadata2	Succeeded	Get Metadata	6/28/2026, 3:10:08 PM	12s	AutoResolveIntegrationRuntime (Southeast